

HYBRID CNN TRANSFORMER APPROACH WITH TRANSFER LEARNING FOR MULTI PLANT DISEASE DETECTION

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ABSTRACT

Accurate plant disease classification is essential for ensuring food security and improving crop yields, especially in agriculture-dependent regions. In this study, we propose a deep learning pipeline for classifying various leaf diseases using a hybrid CNN and Transformer architecture. Our model integrates an Inception V3 backbone with transformer encoder blocks, enabling both local feature extraction and global context modeling. Evaluated on PlantVillage, Cropped PlantDoc and New Plant Diseases dataset, our model achieves a test accuracy of 99.57%, 79.32% and 98.70%. Our work lays the groundwork for future extensions involving segmentation and natural language explanations to improve accessibility for farmers.

1. INTRODUCTION

Agriculture remains the backbone of developing economies, with a significant portion of the population relying on crop production for their livelihood. Traditional manual diagnosis requires expertise and time, which many farmers in developing regions lack. Early and accurate detection of plant diseases can prevent major losses.

Recent advancements in deep learning have shown promise in automating plant disease detection. Convolutional Neural Networks (CNNs) have been extensively utilized for this purpose, offering automated feature extraction and classification capabilities. For instance, Dey et al. (2024) demonstrated the efficacy of pre-trained CNN architectures like AlexNet, VGG16, and VGG19, achieving impressive testing accuracies exceeding 95% on the PlantVillage dataset [1]. Similarly, Harakannanavar et al. (2022) employed machine learning techniques, incorporating image pre-processing methods such as Histogram Equalization and K-means clustering for segmentation, to detect tomato leaf disorders [2].

While these studies highlight the potential of deep learning in plant disease detection, they predominantly rely on the PlantVillage dataset. This dataset comprises images captured under controlled conditions, featuring individual leaves against uniform backgrounds. Such idealized images lack the variability and complexity present in real-world scenarios, limiting the generalizability of models trained solely on this

data. Xu et al. (2024) further emphasize this limitation, noting that while models often achieve near-perfect accuracy on controlled datasets such as PlantVillage, their performance drops sharply on field images since current plant disease datasets “are far from real scenarios” [3]. To address this gap, we evaluate our proposed hybrid CNN–Transformer pipeline not only on PlantVillage but also on datasets that introduce greater variability and realism. In particular, we leverage the Cropped PlantDoc dataset [4], which removes background clutter while still reflecting real-world leaf appearances, and the New Plant Diseases dataset [5], which expands coverage across multiple crops and disease categories under diverse conditions. Singh et al. (2020) introduced the Cropped PlantDoc dataset to address the gap between controlled laboratory images and real-world field conditions, making it more representative of the challenges faced by farmers in practical scenarios.

To address this gap, we propose a hybrid CNN–Transformer pipeline that leverages the localized feature extraction ability of CNNs and the global context modeling power of Transformers. Recently, Vision Transformers have also been explored for plant disease detection. Shandilya et al. (2025) similarly proposed a hybrid CNN–Vision Transformer model for maize leaf disease detection, demonstrating that combining localized CNN features with the global attention capabilities of Transformers significantly improves classification accuracy [6]. Kunduracioğlu and Paçal (2024) applied a Swin-Transformer model to grape leaf classification, achieving near-perfect accuracy on a controlled dataset, though they noted the limited generalizability of such results beyond laboratory conditions [7]. Unlike prior works that focus narrowly on controlled datasets and on specific diseases, we evaluate our model across three different datasets with multiple classes: PlantVillage, Cropped PlantDoc, and Plant Diseases dataset. This comprehensive evaluation allows us to examine both idealized and real-world conditions, highlighting the strengths and weaknesses of the model in diverse scenarios.

Our contributions can be summarized as follows:

- We develop a **hybrid CNN–Transformer architecture** that combines an Inception V3 backbone with transformer encoder blocks to balance local and global feature learning (see Fig. 1).
- We perform extensive experiments on three benchmark

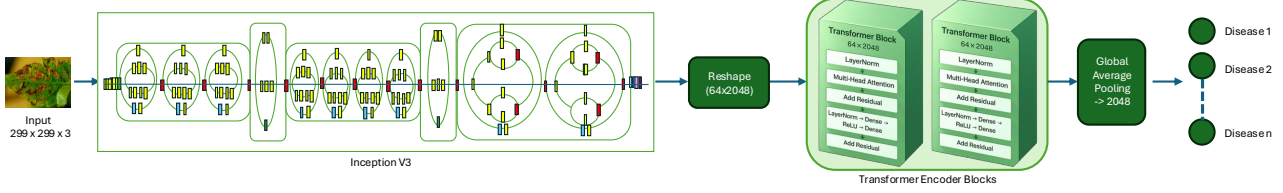


Fig. 1. Proposed hybrid CNN–Transformer architecture. Image is passed through an Inception V3 backbone pre-trained on ImageNet to extract convolutional feature maps. These feature maps are reshaped into a sequence of 64 tokens and forwarded into stacked Transformer encoder blocks, which capture global dependencies across the leaf surface. The encoded sequence is then aggregated with global average pooling, regularized with dropout, and finally classified by a fully connected softmax layer into the target plant disease categories.

multi-class dataset namely PlantVillage, Cropped Plant Doc and New Plant Disease. These datasets represents both controlled and natural settings, demonstrating cross-dataset robustness.

In the subsequent sections, we detail our methodology, present the results of our experiments, discuss the implications of our findings, and outline potential directions for future research.

2. METHODOLOGY

2.1. Model Architecture

The proposed model is designed as a hybrid framework that combines the strengths of convolutional neural networks (CNNs) and Transformers to capture both local and global representations of diseased leaf images. At the front end, we employ an Inception V3 backbone pre-trained on ImageNet. The CNN component is particularly effective at learning localized texture patterns such as spots, blights, and discolorations that are characteristic of specific plant diseases. Schuler et al. (2022) explored this idea of architectural innovation through a dual-branch Inception V3 CNN that separately processed color and luminance channels, achieving 76.91% accuracy on the Cropped PlantDoc dataset [8]. While their method was purely CNN-based, it highlights how specialized architectures can boost performance, providing context for our hybrid CNN–Transformer design. The resulting convolutional feature maps, with dimensions of $8 \times 8 \times 2048$, are then reshaped into a sequence of 64 tokens, each representing a high-dimensional patch embedding.

These sequential embeddings are subsequently passed through two stacked Transformer encoder blocks. Each encoder block comprises a multi-head self-attention mechanism followed by feedforward layers and residual connections, enabling the model to capture long-range dependencies and contextual relationships across different parts of the leaf surface. Finally, the output sequence is aggregated using a global average pooling layer, which reduces the representation to a compact feature vector. This vector is regularized with dropout to prevent overfitting and is passed through a fully connected softmax classifier that produces the final

probability distribution over the disease categories. Fig. 1 represents the proposed architecture.

2.2. Training Strategy

We train the model in three stages:

1. **Stage 1:** Freeze CNN layers, train Transformer and classifier head.
2. **Stage 2:** Unfreeze CNN partially, fine-tune with lower learning rate.
3. **Stage 3:** Fully unfreeze CNN for joint training.

3. EXPERIMENTAL SETUP

3.1. Datasets

We evaluate on three datasets with varying levels of difficulty. Table 1 provides an overview of the three datasets in this study.

- **PlantVillage:** A laboratory dataset of approximately 54,303 leaf images with plain backgrounds and 38 disease/crop classes.
- **Cropped PlantDoc:** Bounding box–based crops from PlantDoc, designed to reduce background noise.
- **New Plant Diseases Dataset:** A custom dataset with 38 classes and around 87,000 images of healthy and diseased crop leaves.

3.2. Implementation Details

For model training, we used categorical cross-entropy as the loss function and optimized the network using Adam with a learning rate decay schedule, starting from an initial learning

Table 1. Dataset statistics: number of images, classes, resolution, train/test splits, and training epochs.

Dataset	Images	Classes	Resolution	Split	Epochs
PlantVillage [1]	54,303	38	256×256	80/20	15
Cropped PlantDoc [4]	9,216	27	224×224	65/15/20	200
New Plant Diseases [5]	~ 87,000	38	224×224	80/20	35

rate of 1×10^{-4} . To reduce overfitting and address class imbalance, we employed dropout with a rate of 0.3 along with class weighting. Data augmentation strategies included random flips and rotations, and we further applied test-time augmentation (TTA) to enhance robustness. The models were trained with a batch size of 32.

4. RESULTS

4.1. Overall Performance

Tables 2, 3, and 4 summarize the classification accuracies of our hybrid model on the three datasets. As expected, Cropped PlantDoc has a lower accuracy as compared to PlantVillage and New Plant Diseases; however, it remains comparable to the results reported in other studies.

Table 2. Comparison of classification accuracies achieved by different deep learning and machine learning architectures on the PlantVillage dataset [9]. The results include models trained from scratch as well as those employing transfer learning. The reported accuracies highlight the effectiveness of transfer learning with our proposed hybrid model achieving the highest accuracy of 99.57%, outperforming previously reported methods.

Architecture	Test Accuracy (%)
GoogleNet (Learning from Scratch)	94.31
GoogleNet (Transfer Learning)	98.21
AlexNet (Learning from Scratch)	91.30
VGG16	97.23
AlexNet	97.49
SVM + K-Means + ANN	95.71
AlexNet (Transfer Learning)	99.34
Residual Teacher-Student CNN	99.05
K-Means + GLCM + SVM	93.50
MobileNetV2	95.70
Our Results	99.57

Table 3. Accuracy comparison on the Cropped PlantDoc dataset [8], using different combinations of the CIELAB color space channels. The experiments combine 20% luminance (L) with 80% chromaticity (a/b) features across varying layer depths. While the 6-layer configuration achieves the best reported accuracy of 77.08%, our proposed model further improves performance to 79.32%.

Architecture	Test Accuracy (%)
20%L + 80%AB (1 layer)	72.50
20%L + 80%AB (2 layers)	76.97
20%L + 80%AB (4 layers)	74.68
20%L + 80%AB (6 layers)	77.08
20%L + 80%AB (8 layers)	73.90
20%L + 80%AB (10 layers)	73.84
20%L + 80%AB (11 layers)	74.23
Our Results	79.32

Table 4. Accuracy comparison on the New Plant Diseases dataset [5]. Conventional CNN and MobileNetV2 models achieve accuracies of 96.44% and 96.89%, respectively, while EfficientNetB0 reports the highest previously published accuracy of 99.69%. Our proposed hybrid model attains a second highest performance of 98.70%, clearly outperforming the baseline CNN and remaining competitive with state-of-the-art methods.

Architecture	Accuracy (%)
CNN	96.44
MobileNetV2	96.89
EfficientNetB0	99.69
ResNet-50	79.83
Our Results	98.70

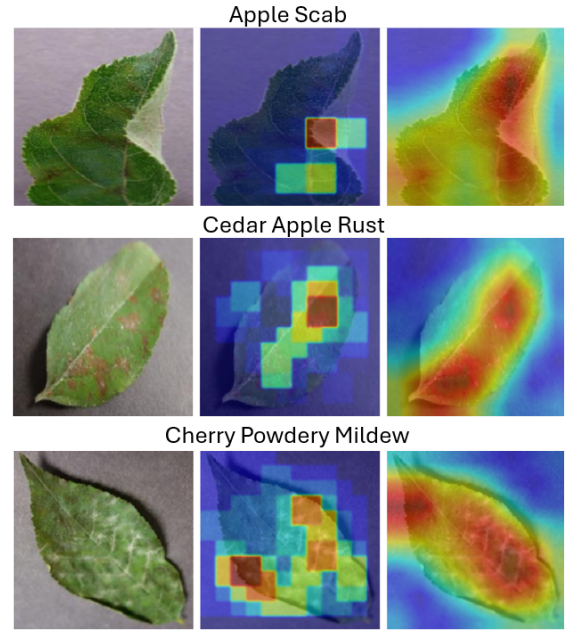


Fig. 2. Visualization of model decision-making for plant disease diagnosis on Plant Village. Left: Original image, Center: Occlusion sensitivity analysis (red = high impact), Right: Spatial feature importance from CNN backbone.

To further interpret the model’s decision-making process, we generated some visualizations on the PlantVillage samples. Fig. 2 shows some visualizations. Each example shows the raw leaf image alongside two interpretability overlays: an occlusion map, which highlights performance drops when certain regions are masked, and an attention map, which reveals where the model focuses during classification during the early stages. Together, these visualizations confirm that the model consistently attends to lesion areas rather than background noise.

4.2. Dataset-wise Discussion

PlantVillage. Our hybrid model achieves an accuracy of 99.57% on the PlantVillage dataset. This near-perfect performance is consistent with previous CNN-based studies, such as Mohanty et al. (2016), who reported accuracies exceeding 99% under controlled conditions using deep CNN architectures [10].

Cropped PlantDoc. When background clutter is removed via segmentation, accuracy improves dramatically to 79.32%. This not only surpasses the usual CNN baselines reported in the 60–70% range, but also exceeds the previously best-known result of 76.91% using a two-branch CNN [8]. Schuler et al. (2022) established this benchmark with a dual-branch InceptionV3 architecture that processed color and luminance channels separately, making our result of 79.32% a new state-of-the-art on Cropped PlantDoc. To the best of our knowledge, this establishes a new state of the art on Cropped PlantDoc.

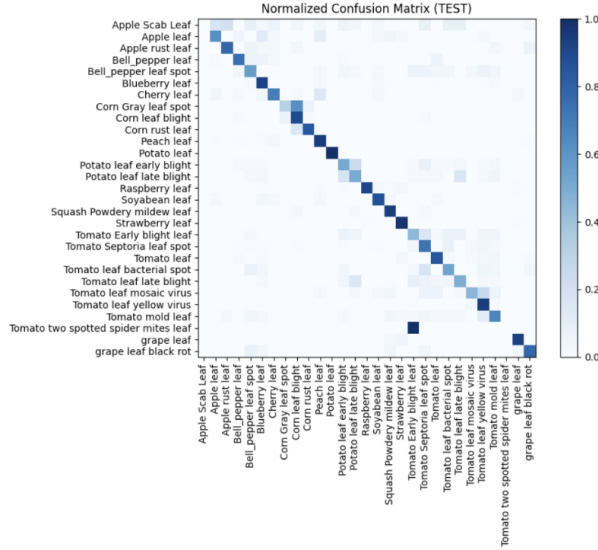


Fig. 3. Normalized confusion matrix on the Cropped PlantDoc test set, showing class-wise prediction performance across multiple plant species and diseases. The anomaly for Tomato two spotted spider mites leaf is due to the low training sample of the disease.

New Plant Diseases Dataset. On the New Plant Diseases dataset, our model achieves 98.70% accuracy. This dataset is essentially an augmented extension of PlantVillage, recreated offline to expand the number of samples while maintaining class diversity. Despite the scale-up, our model’s accuracy remains consistent with PlantVillage results and even surpasses several reported CNN baselines in prior work. Nigar et al. (2023) similarly reported strong performance on this dataset, achieving high classification accuracy by employing a deep-learning prediction model with explainable AI, further vali-

dating its utility for benchmarking advanced architectures [5]. This consistency demonstrates that our hybrid approach generalizes well even when exposed to a larger, more diverse augmented dataset.

In summary, these results show that our method is not only comparable to—but in some cases even stronger than—several prior approaches across different datasets. In simpler datasets, it keeps pace with the best baselines, and in tougher, cluttered environments it delivers noticeable improvements. In plain terms, this suggests that the approach is robust, scales well with larger and augmented datasets, and provides a solid foundation for further innovation.

5. CONCLUSION AND FUTURE WORK

We proposed a hybrid CNN–Transformer pipeline for plant disease classification and evaluated it on three benchmark datasets. The model achieved strong results on controlled datasets and showed improved generalization to real-world settings compared to CNN-only baselines. These findings demonstrate the potential of combining convolutional and transformer-based features for more reliable agricultural applications.

Future work will focus on integrating leaf segmentation to reduce background noise, as well as exploring self-supervised pretraining for better robustness in diverse field conditions. We also aim to provide natural-language explanations to improve usability for farmers and to test lightweight versions of the model for mobile deployment. Prior studies, such as Ramcharan et al. (2019) with the PlantVillage Nuru app [11] and Askale et al. (2025) on maize disease detection [12], highlight the importance and feasibility of on-device, real-time diagnosis.

6. REFERENCES

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